

DUET: Endogenous Failure Detection and Recovery via Dual-Action Manifold Alignment

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Abstract—Vision-language-action (VLA) models map vision and language to robot actions. With limited post-training demonstrations, however, they often cannot detect or recover from execution failures that move the robot out of distribution (OOD), such as mis-grasps, dropped objects, unexpected occlusions, and object or robot displacement. Existing solutions rely on external failure detectors and costly intervention or recovery data, leaving detection loosely coupled with control. We introduce DUET (Dual-manifold Unified Execution and Transfer), a dual-action-head module that predicts relative and absolute actions. DUET uses the endogenous disagreement between the two heads as its primary failure signal, complemented by state-distribution distance. The absolute head excludes the current robot state and predicts RVQ-anchored global waypoints from vision and language, providing stable recovery targets when relative control becomes unreliable. We evaluate DUET in LIBERO simulation and real-world manipulation under diverse failure events. DUET preserves nominal success (TODO% vs. TODO% for the relative-only baseline) while achieving TODO% failure recovery success, outperforming the strongest baseline by TODO percentage points.

I. INTRODUCTION

Vision-language-action (VLA) models have recently shown strong capability for general-purpose robot manipulation [1], [2]. In real-world deployment, however, robots still encounter situations outside the training distribution that cascade into execution failure and limit robustness. Prior work typically organizes such generalization problems into *semantic*, *visual*, and *behavioral* distribution shifts [3].

In a typical VLA loop, a vision-language model f_{VLM} encodes observations, language, and robot state into task-relevant representations; an action policy then produces control commands. Most efforts to improve generalization concentrate on this first stage—enlarging the pretrained backbone, broadening vision-language pretraining, or collecting more robot and visual data—and thereby strengthen the visual-semantic representation of f_{VLM} . Such methods can mitigate semantic OOD and much of visual OOD, but they usually incur larger models and extra data-collection cost. This raises a natural question: *if the base VLM and the available training data are held fixed, where else can VLA OOD robustness still be improved?*

We argue that an underexplored source of robustness lies in how robot behavior is represented and executed. This direction is especially relevant to behavioral OOD, where the robot may already understand the task, yet its own state, the world state, or the outcome of action execution has departed from the successful behavior seen during training.

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For transient visual failures such as short camera blackouts or dropped frames, the action/control layer may also provide execution continuity while vision is temporarily unavailable.

Existing OOD and failure detectors typically add a monitor, verifier, or uncertainty module outside the policy, or collect extra failure and correction data [4], [5], [6], [7], [8]. Yet OOD and failure are not equivalent: an unseen state may still be handled successfully, while a policy can also fail in distribution through prediction or compounding control error. For closed-loop manipulation, a more practical question than “has the current step already failed?” is therefore: *can the robot notice that its current execution is beginning to depart from behavior that can still complete the task?* We seek this signal endogenously from the policy’s own action predictions, rather than from an extra external detector.

Action representation offers a natural entry point. Current VLA policies usually adopt a single representation, most often relative actions [9], [10]. A relative action describes how to continue from the current state and is well suited to fast, fine-grained local control, but because it depends on that state it provides no persistent global reference once the robot is perturbed, an object is displaced, or execution error accumulates—and thus cannot tell whether motion is still heading toward the intended goal. An absolute action, by contrast, predicts a future end-effector pose that remains a global task-space reference and can still indicate a recovery direction after the robot state has shifted. The two representations therefore have complementary control properties.

This leads to a further question: *can complementary relative and absolute actions be used jointly so that a VLA can both sense abnormal execution drift and recover from it, without modifying the base VLM or relying on extra failure-recovery data?* Doing so requires keeping the two representations complementary rather than collapsing into similar predictors; constructing a reliable execution-inconsistency signal from the two streams without OOD or failure labels; and turning that signal into continuous recovery control rather than a binary failure flag.

We propose **DUET**, a dual-action framework for robust VLA execution. DUET keeps the original vision-language backbone unchanged and jointly learns a state-aware relative stream and a state-independent absolute stream. Relative actions provide local fine-grained control under nominal conditions; absolute actions supply a global task-space anchor. Transporting each absolute prediction into the current robot frame lets DUET compare the two judgments of the current motion direction. As they diverge, the disagreement characterizes inconsistency with successful behavior

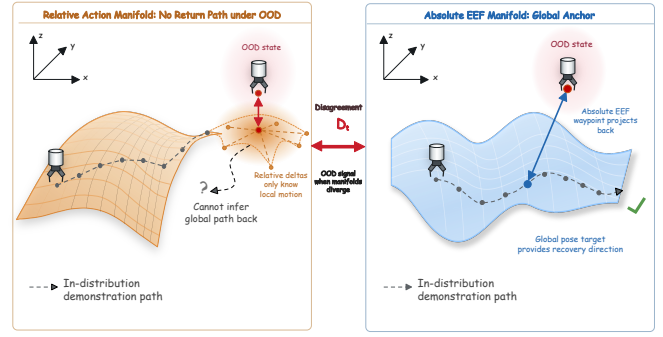
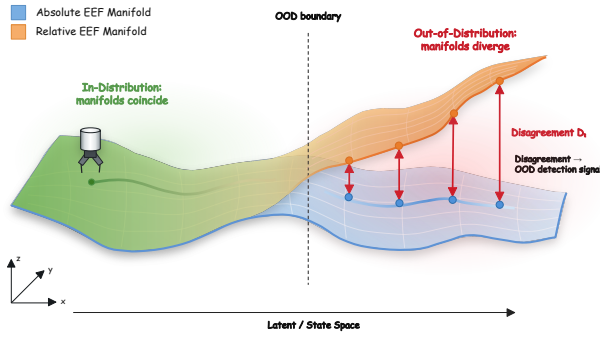


Fig. 1. Dual-manifold failure detection and recovery. The relative and absolute end-effector manifolds agree during nominal execution but diverge when a failure drives the system OOD, exposing an endogenous disagreement score D_t . Relative actions provide precise local motion, whereas absolute waypoints project execution toward a stable demonstrated-manifold anchor.

and dynamically reweights relative control against absolute correction. The same pair therefore answers *when* execution is becoming unreliable and *where* recovery should proceed. Under transient visual failure, the most recent reliable absolute waypoint can further serve as a short-lived action reference, improving execution continuity.

We evaluate DUET under robot-state perturbations, environment/object-state changes, execution-dynamics shifts, and transient visual observation failures, testing whether the action/control layer alone can further improve VLA deployment robustness. Our contributions are as follows:

- We revisit VLA OOD robustness from the action and control layer, and show that, with the base VLM and existing training data held fixed, action representation itself can strengthen robustness under behavioral OOD and transient visual failure.
- We propose DUET, which builds complementary dual-action representations from state-aware relative actions and state-independent absolute actions. Their cross-representation disagreement provides an endogenous execution-inconsistency signal, while the absolute stream supplies the global reference needed for recovery.
- We design an adaptive dual-action closed-loop controller and evaluate it under robot-state, world-state, execution-dynamics, and transient visual perturbations, showing that DUET improves robust execution without an extra failure detector or dedicated recovery data.

II. RELATED WORK

A. Failure Detection for Robot Policies

Runtime failure detection estimates whether a policy remains within the support of successful behavior. Diff-Dagger uses diffusion uncertainty to request expert intervention [11], while Sentinel monitors action inconsistency and task progress [12]. FAIL-Detect formulates failure detection as sequential OOD testing using only successful executions [13]; FIPER similarly combines representation novelty with action uncertainty without failure data [14].

VLA-specific methods use internal features or external foundation models. SAFE learns a detector from successful

and failed rollouts [15], whereas AHA trains a VLM to detect and diagnose failures [16]. Such methods require additional supervision or a separate detector whose output is not a recovery action. DUET instead detects failures through disagreement between two action spaces learned within the policy, using OOD manifold inconsistency as both an alarm and a recovery signal.

B. Recovery from Execution Failures

Recovery methods commonly expand the training distribution with expert corrections. DAGger queries experts on policy-induced states [17]; intervention-based methods collect human takeovers [18], [19]; and IntervGen transforms limited interventions into corrective trajectories [20]. These approaches reduce compounding errors but still require additional recovery actions.

Foundation-model systems instead use semantic supervisors. RACER learns from language-annotated recovery trajectories [21], while AHA and VLM controllers diagnose failures and guide downstream correction [16], [22]. Their detection and recovery components remain separately trained or prompted. DUET couples both capabilities without extra recovery data: absolute states from successful demonstrations provide recovery anchors, while cross-space disagreement determines when and how strongly to move toward them.

III. METHOD

A. Problem Formulation

At time t , the robot observes multi-view images o_t , a language instruction l , and its current proprioceptive state s_t . DUET is trained only on successful demonstrations and predicts the demonstrated behavior in two action representations over a horizon K :

$$\hat{\Delta}A_t = f_{\text{rel}}(o_t, l, s_t) \in \mathbb{R}^{K \times 7}, \quad (1)$$

$$\hat{W}_t = f_{\text{abs}}(o_t, l) \in \mathbb{R}^{K \times 7}. \quad (2)$$

Here, ΔA_t contains relative operational-space-control commands and W_t contains future absolute end-effector waypoints $[x, y, z, r_x, r_y, r_z, g]$. We decompose each command as $a_t = [a_{p,t}, a_{g,t}]$, where $a_{p,t} \in \mathbb{R}^6$ controls end-effector pose and $a_{g,t} \in \mathbb{R}$ controls the gripper. A successful

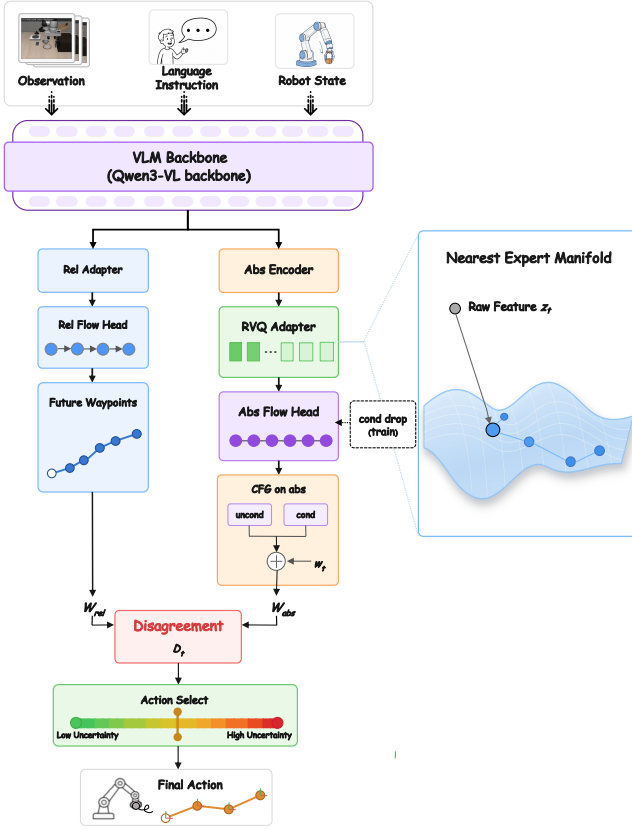


Fig. 2. DUET in four stages: a shared vision-language encoder, relative and absolute action heads, cross-space failure detection, and closed-loop recovery. The relative head provides state-aware local control; the explicitly state-free absolute head uses RVQ-anchored conditions to predict global waypoints. Their disagreement determines the transition between nominal execution and recovery.

trajectory naturally supplies both labels: its recorded delta actions supervise f_{rel} , while its future end-effector states supervise f_{abs} . The geometric analysis and fusion operate on a_p ; although the absolute head also predicts gripper state during training, the deployed a_g always comes from the relative head.

B. Dual Action Manifolds

For an instruction l , let $\mathcal{M}_l \subset SE(3)$ denote the end-effector state manifold traced by successful demonstrations, and let $T\mathcal{M}_l$ denote its tangent bundle. Under nominal execution, the relative head predicts a local motion vector

$$\hat{\delta}a_{p,t} = f_{\text{rel}}(o_t, l, s_t)_p \in T_{p_t}\mathcal{M}_l, \quad (3)$$

where p_t is the current end-effector pose. This tangent representation is locally smooth and expressive, but it specifies only how to move from the current chart. Once a failure displaces the robot from the demonstrated support, the extrapolated vector field contains no explicit destination to which the robot should return.

The absolute head instead predicts a future coordinate

$$\hat{w}_{p,t} = f_{\text{abs}}(o_t, l)_p \in \mathcal{M}_l, \quad (4)$$

which acts as a visual-language-conditioned global anchor. To compare the two representations, we introduce a state-dependent transport map

$$c_t = \Psi_{p_t}(\hat{w}_{p,t}) \in T_{p_t}SE(3), \quad (5)$$

which converts the absolute waypoint into a local correction at the live pose. Along an in-domain trajectory, c_t and $\hat{\delta}a_{p,t}$ describe the same demonstrated motion in two coordinate systems and should therefore align. When a failure drives execution OOD, the relative head extrapolates a local field while the absolute head remains anchored to a demonstrated destination, exposing a cross-space inconsistency.

This geometry couples detection and recovery without asserting a formal OOD guarantee: the norm of the inconsistency provides a scalar failure signal, whereas the transported vector c_t already provides the corresponding recovery direction. DUET can therefore change both when and where the controller moves using the same pair of predictions.

C. Dual-Head Architecture

Figure 2 shows the shared VLM and two flow-matching action heads [23], [24]. The relative head attends to the full hidden sequence and explicitly receives s_t through a numerical state encoder. It is therefore state-aware and serves as the primary controller during nominal execution. The absolute head receives no explicit current proprioception. When state tokens are appended to the causal VLM prompt, they are placed last and excluded from absolute-branch pooling, while the relative head can still attend to them. This separation prevents the absolute predictor from collapsing into an integration of the current state and preserves its role as a global reference.

The absolute branch pools its state-free hidden sequence H^0 and applies residual vector quantization (RVQ) [25], [26]:

$$Z_q = \text{RVQ}(\text{Pool}(H^0)). \quad (6)$$

RVQ maps continuous conditions to a discrete atlas learned from successful demonstrations, anchoring absolute predictions to recurrent visual-language modes rather than an unconstrained feature vector. Its residual norm is retained only as auxiliary telemetry and is not part of the primary failure score. The absolute flow head additionally uses classifier-free guidance (CFG) [27], enabling its conditional prediction to be softened toward the demonstrated waypoint marginal during recovery.

D. Disagreement-Preserving Learning

The two heads are supervised by the paired labels available in every successful trajectory. Both use flow-matching objectives, while RVQ contributes a commitment loss. The complete default objective is

$$\mathcal{L} = \lambda_{\text{rel}}\mathcal{L}_{\text{rel}} + \lambda_{\text{abs}}\mathcal{L}_{\text{abs}} + \lambda_{\text{vq}}\mathcal{L}_{\text{vq}}, \quad (7)$$

where the absolute labels are normalized future states and the relative labels are the recorded delta commands. We

deliberately impose no cross-head consistency loss or distillation: forcing agreement everywhere would suppress the inconsistency needed for failure detection.

DUET instead specializes the branches through asymmetric visual augmentation. The absolute head and RVQ adapter train on both clean and perturbed observations, encouraging the global anchor to remain stable under degraded input. The relative head is optimized only on the clean subset $\mathcal{B}_{\text{clean}}$:

$$\mathcal{L}_{\text{rel}} = \frac{1}{|\mathcal{B}_{\text{clean}}|} \sum_{i \in \mathcal{B}_{\text{clean}}} \ell_{\text{FM}}(\hat{\Delta}a^{(i)}, \Delta a^{(i)}). \quad (8)$$

This gradient truncation preserves relative-head sensitivity while training the absolute branch as the robust reference, making cross-space disagreement more discriminative under visual OOD. For CFG training, the absolute head also replaces its quantized condition with learned null tokens on a subset of samples and learns both conditional and marginal waypoint flows.

E. Joint Failure Detection and Recovery

At deployment, the policy returns both unnormalized action chunks. Let the live pose be $p_t = (x_t, R_t)$ and the pose component of the current absolute waypoint be $\hat{w}_{p,t} = (\hat{x}_t, \hat{R}_t)$. DUET implements the transport map in Eq. (5) as

$$c_t = \text{clip}\left(\left[\frac{(\hat{x}_t - x_t)/\sigma_p}{\text{Log}(\hat{R}_t R_t^{-1})/\sigma_R}\right], -1, 1\right), \quad (9)$$

where $\text{Log} : SO(3) \rightarrow \mathfrak{so}(3)$ gives the geodesic rotation correction and σ_p, σ_R convert pose error into action units. The cross-space disagreement is

$$D_t = \frac{\|c_t - \text{clip}(\hat{\delta}a_{p,t}, -1, 1)\|_2}{\sqrt{6}}. \quad (10)$$

This is the primary failure signal. We supplement it with a coarse state-distance term that measures normalized violation of the training-state bounds:

$$S_t = \text{mean} \left[\frac{\max(s_{\min} - p_t, 0)}{s_{\max} - s_{\min}} + \frac{\max(p_t - s_{\max}, 0)}{s_{\max} - s_{\min}} \right]. \quad (11)$$

The resulting failure score and recovery weight are

$$q_t = D_t + \lambda S_t, \quad (12)$$

$$\alpha_t = \sigma(k(q_t - \tau)). \quad (13)$$

DUET then interpolates between local execution and global recovery while retaining relative gripper control:

$$a_t = [a_{p,t}, a_{g,t}], \quad (14)$$

$$a_{p,t} = (1 - \alpha_t)\hat{\delta}a_{p,t} + \alpha_t c_t, \quad (15)$$

$$a_{g,t} = \hat{\delta}a_{g,t}, \quad (16)$$

where $[\hat{\delta}a_{p,t}, \hat{\delta}a_{g,t}]$ is the relative-head command. Thus relative control dominates along the demonstrated support, while a rising failure score shifts a_p toward the absolute recovery direction; control returns smoothly after the two representations realign. When requesting a new chunk, the

previous fusion weight also softens the absolute-head CFG scale,

$$w_t = w_{\max} - (w_{\max} - w_{\min})\alpha_{t-1}. \quad (17)$$

The disagreement D_t therefore determines *when* recovery is needed, while the vector c_t determines *how* to recover.

IV. EXPERIMENTS

A. Experimental Questions and Protocol

Our experiments answer three questions in order. **Q1: Nominal capability.** Does the absolute branch preserve standard LIBERO performance relative to established VLA baselines and the relative branch of the same checkpoint? **Q2: OOD robustness.** Does adaptive dual-action control improve closed-loop completion under behavioral and transient visual shifts? **Q3: Component necessity.** Which training and inference components are responsible for the gain?

We use the four LIBERO suites [28]—Spatial, Object, Goal, and LIBERO-10—with 10 tasks per suite and 50 initial states per task (500 episodes per method per suite). Each DUET suite uses a separate dual-head checkpoint trained for 10k steps. For OOD experiments, *DUET-Rel* executes only the relative head of that same checkpoint, whereas *DUET* uses adaptive fusion. This is the primary causal comparison: the VLM, training data, checkpoint, task, initial state, and perturbation are held fixed, and only use of the absolute branch changes.

Perturbations are injected at a fixed control step using deterministic event seeds, so DUET and DUET-Rel are paired on identical (task, init) keys. We report task success, paired McNemar p -values, and injection coverage. When an event requires a semantic precondition—a grasp for object drop, for example—we additionally report results among episodes in which both methods received the perturbation; untriggered episodes are never silently discarded from the overall rate.

The bound-violation term S_t is almost always zero on LIBERO. For robot-state and world-state shifts we therefore scale disagreement by the residual to the current absolute waypoint,

$$q_t = D_t \cdot \text{clip}(\|c_{t,xyz}\|/0.05, 0, 1), \quad (18)$$

so the gate rises only when disagreement coincides with a meaningful correction distance. Action-execution and contact experiments use the default score $q_t = D_t + \lambda S_t$.

B. Standard LIBERO Benchmark and Baselines

Table I separates two roles of a baseline. Reported $\pi_{0.5}$ and StarVLA-GR00T scores position DUET against standard VLA systems; they do not isolate the dual-action design because their training mixtures and optimization budgets differ. DUET versus DUET-Rel is the controlled within-checkpoint comparison that tests whether the additional branch harms nominal execution. Full clean runs for these two rows are pending and are not inferred from OOD or smoke-test episodes.

TABLE I

STANDARD LIBERO SUCCESS (%). EXTERNAL ROWS ARE REPORTED RESULTS; STARVLA USES JOINT 30K TRAINING, WHEREAS DUET USES SUITE-SPECIFIC 10K CHECKPOINTS.

Method	Spatial	Object	Goal	LIBERO-10	Avg.
$\pi_{0.5}$ (reported)	98.8	98.2	98.0	92.4	96.9
StarVLA-GR00T (reported)	97.8	98.8	97.4	92.0	96.5
DUET	98.8	99.4	97.6	91.8	96.9
DUET-rel	98.0	99.2	92.0	86.0	93.8

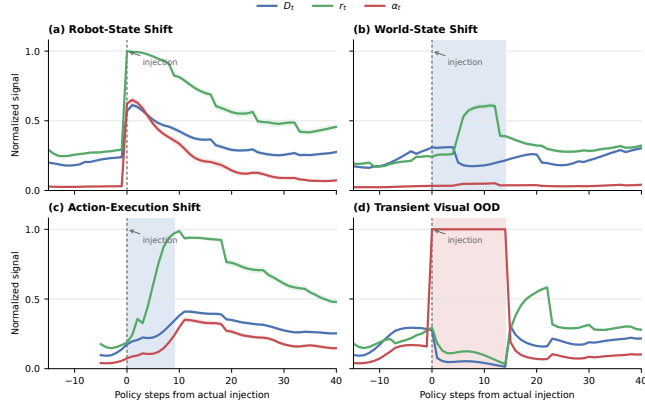


Fig. 3. Aggregate DUET dynamics for four OOD protocols on the Object suite: (a) end-effector teleport, (b) vertical grasp drop, (c) action perturbation, and (d) camera blackout with a cached waypoint. Traces are aligned to the actual injection step and show the mean with pointwise bootstrap 95% confidence intervals over successfully injected episodes. Shading marks the perturbation or forced-hold window.

C. OOD Robustness

The OOD main tables intentionally contain only DUET and DUET-Rel. Original StarVLA may be added as a secondary reference only after rerunning it with identical initial states, injection steps, severities, directions, and success accounting; $\pi_{0.5}$ is treated the same way. Until then, mixing reported clean scores with the paired OOD study would not provide a valid causal baseline.

1) *OOD Taxonomy*: We organize OOD by the locus at which successful execution support is violated (Table II).

Behavioral OOD keeps vision usable but perturbs one of three elements of closed-loop behavior. **Robot-state shift** changes the robot pose directly. **World-state shift** changes the manipulated object or grasp while the robot may remain near its old pose. **Action-execution shift** changes the map from the policy command to the executed command. These loci cover robot state, world state, and the action channel without conflating action corruption with physical robot-world contact.

Transient visual OOD instead removes observations for a short horizon. The dual heads must not be queried on black frames; DUET holds the last reliable absolute waypoint until vision returns. Contact force is reported after the behavioral experiments as an additional interaction stress test because it changes contact dynamics but often leaves no persistent waypoint residual.

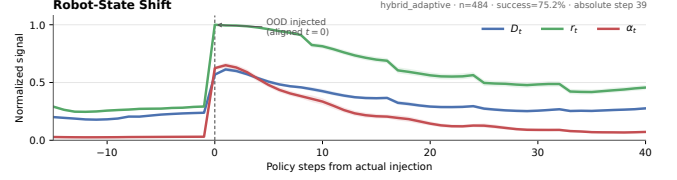
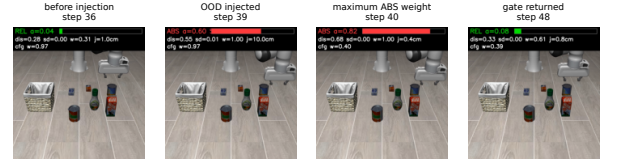


Fig. 4. Robot-state shift (end-effector teleport) on the Object suite. Top: four frames from a successful episode, showing relative control before the jump, an immediate switch to absolute recovery after the arm is displaced, and return to relative control once the pose residual collapses. Bottom: mean D_t , residual weight $r_t = \text{clip}(\|c_{t,xyz}\|/0.05, 0, 1)$, and α_t over 484 successfully injected episodes, aligned at the actual injection step, with pointwise bootstrap 95% confidence intervals. Shading marks the perturbation window.

2) Behavioral OOD:

a) *Robot-State Shift*: An instantaneous end-effector teleport leaves the arm off the demonstrated chart while objects typically remain in place. The residual-weighted score $q_t = D_t \cdot r_t$, with $r_t = \text{clip}(\|c_{t,xyz}\|/0.05, 0, 1)$, is therefore appropriate: the demonstrated grasp target remains valid, so a large waypoint residual should raise the gate rather than be treated as a stale world reference.

Figure 4 shows the intended mechanism. The top strip is one successful Object episode: $\alpha_t \approx 0$ (relative control) immediately before injection, a switch to absolute recovery as soon as the arm is displaced, and a return to relative control after the residual collapses. The lower panel aggregates 484 successfully injected Object episodes aligned at the actual injection step. D_t and r_t jump at $t=0$, α_t rises from near zero to above 0.6, and both decay as the arm returns. The same aggregate pattern appears in Fig. 3(a). Table III shows that DUET raises overall success from 40.0% to 61.7% (+21.7 pp over 2000 paired episodes). The gain is concentrated where a valid grasp target still exists after the jump: Object +39.8 pp, LIBERO-10 +25.0 pp, Goal +19.0 pp. After successful injection, Object recovery is 72.8% for DUET versus 33.2% for DUET-Rel, consistent with a large residual $\|c_t\|$ unblocking disagreement-driven return.

Spatial is not significant (+3.0 pp, $p=0.21$): about 12% of episodes die before the scheduled step, and several Spatial tasks already operate near cluttered supports where a 5 cm jump is a smaller relative error. Goal remains weak on opening the top drawer and on pushing a plate (1–3/50 and –18 pp, respectively), i.e., contact-rich primitives whose demonstrated waypoints are poor recovery targets.

b) *World-State Shift*: The revised grasp-drop protocol models a failed hold rather than a lateral knock. At policy step 67, if a free-joint object is in gripper contact, the

TABLE II

OOD FAMILIES AND SHIFT LOCI. THE ABSOLUTE WAYPOINT IS USEFUL ONLY WHEN THE DEMONSTRATED WORLD REFERENCE REMAINS VALID.

Family	Shift	Protocol	What changes	Absolute waypoint
Behavioral	Robot-state	EEF teleport	Arm pose	Valid return target
	World-state	Vertical grasp drop	Object / grasp	Potentially stale
	Action-execution	Action perturb	Executed a_t	Valid return target
Visual	Transient dropout	Camera blackout	Observation	Hold last waypoint

TABLE III

CLOSED-LOOP SUCCESS (%) ON BEHAVIORAL OOD. DUET-REL AND DUET SHARE A CHECKPOINT. TELEPORT AND VERTICAL DROP USE RESIDUAL-WEIGHTED DISAGREEMENT; ACTION PERTURBATION USES THE DEFAULT DISAGREEMENT GATE.

Setting	Suite	DUET-Rel	DUET	Δ	p
<i>Robot-state shift</i> (end-effector teleport, medium, $N=500$)					
	Spatial	54.6	57.6	+3.0	0.21
	Object	36.2	76.0	+39.8	1×10^{-40}
	Goal	33.4	52.4	+19.0	2×10^{-12}
	LIBERO-10	35.8	60.8	+25.0	1×10^{-17}
	Overall	40.0	61.7	+21.7	—
<i>World-state shift</i> (vertical grasp drop, Object, $N=500$)					
	Object	89.2	95.2	+6.0	2.4×10^{-5}
<i>Action-execution shift</i> (action perturbation, step 5, $N=500$)					
medium	Spatial	54.0	74.0	+20.0	1×10^{-14}
medium	Object	48.8	84.4	+35.6	1×10^{-34}
medium	Goal	47.2	74.2	+27.0	5×10^{-22}
medium	LIBERO-10	36.6	78.6	+42.0	1×10^{-48}
severe	Spatial	12.8	43.2	+30.4	6×10^{-31}
severe	Object	7.8	60.6	+52.8	1×10^{-70}
severe	Goal	10.6	52.6	+42.0	3×10^{-49}
severe	LIBERO-10	5.4	49.0	+43.6	2×10^{-53}
	Overall	27.9	64.6	+36.7	—

evaluator opens the fingers, preserves object XY , lowers the object by 5 cm in Z , and forces the gripper open for 15 steps. The arm itself is not displaced. If no object is grasped, the event is marked `not_grasped` and the simulator is left unchanged.

This protocol tests a different boundary from robot teleport: the robot remains near the demonstrated grasp pose while the manipulated object changes state. In the matched Object run, DUET succeeds on 476/500 episodes (95.2%) versus 446/500 (89.2%) for DUET-Rel, a +6.0 pp gain (bootstrap 95% CI [3.4, 8.8] pp; McNemar $p=2.4 \times 10^{-5}$). Injection succeeds in 498/500 DUET and 494/500 DUET-Rel episodes. Figure 3(b) shows that r_t rises during the forced-open window while mean α_t remains low: the controller preserves local visual re-acquisition rather than blindly following a potentially stale absolute grasp.

A Goal-suite fixed-step diagnostic further shows why injection coverage must be explicit. At step 67, DUET and DUET-Rel both receive the vertical drop in only 282/500 paired episodes because drawer, stove-switch, and rack tasks often have no grasp at that time. Across all episodes the rates are 53.6% and 50.4%, respectively; among both-injected

TABLE IV

EXTERNAL-FORCE STRESS TEST (FIXED-STEP, 100% INJECTION, $N=500$ PER CELL). NOT GROUPED WITH END-EFFECTOR TELEPORT.

Suite	Medium			Severe		
	DUET-Rel	DUET	Δ	DUET-Rel	DUET	Δ
Spatial	96.6	98.4	+1.8	96.8	96.8	0.0
Object	97.6	99.0	+1.4	96.2	96.8	+0.6
Goal	89.2	96.8	+7.6	87.2	96.8	+9.6
LIBERO-10	87.8	92.6	+4.8	86.4	90.4	+4.0
Overall	DUET-Rel 92.22		DUET 95.95	$\Delta = +3.73$		

pairs they are 23.4% and 17.7%. The difference is dominated by one bowl-to-stove task, so we do not treat it as a suite-level World-state result.

c) Action-Execution Shift.: Corrupting the 6D command for a short window drives the arm off the successful trajectory without an explicit pose reset. Unlike teleport, both D_t and α_t rise gradually as execution error accumulates (Fig. 3(c)). This is the strongest dual-action setting in Table III: over eight suite $\times$ severity groups (4000 pairs), DUET attains 64.6% versus 27.9% for DUET-Rel (+36.7 pp). Gains grow with severity (Object severe +52.8 pp). The response is generated by cross-action disagreement, not an external jump detector.

3) Additional Interaction Stress Test.: An external force during contact perturbs robot-world dynamics without explicitly resetting the end-effector pose. Table IV reports eight suite $\times$ severity groups: DUET reaches 95.95% versus 92.22% for DUET-Rel (+3.73 pp). The gain is non-negative in every cell and is largest on Goal (+7.6 and +9.6 pp), where drawer and plate interactions amplify short contact disturbances. The margin is smaller than for teleport or direct action corruption, consistent with force often producing a shorter-lived residual. We therefore report it separately as an interaction stress test rather than relabeling it as either Robot-state or Action-execution shift.

4) Transient Visual OOD.: We blank both cameras at a fixed step. Querying the dual heads on black frames collapses the absolute condition and is harmful (37.7% without a visual guard vs. 73.5% for DUET-Rel on an eight-group step-23 protocol). DUET instead caches the last reliable absolute waypoint and executes its correction with $\alpha=1$ for the dropout horizon.

Figures 3(d) and 5 verify that the controller holds the cached waypoint only during the blackout and resumes its

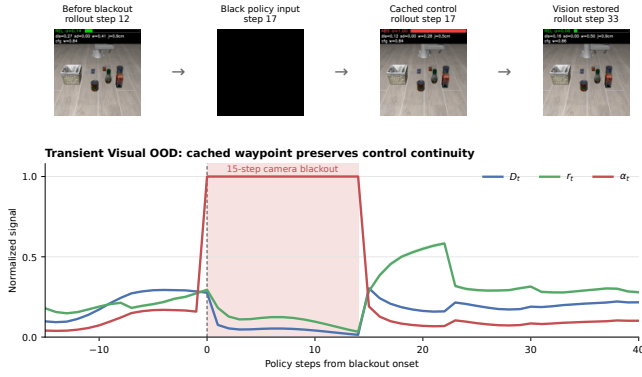


Fig. 5. Camera-blackout mechanism. DUET detects the invalid observation, skips dual-head inference on black frames, executes the correction to the last reliable absolute waypoint with $\alpha_t = 1$, and resumes the adaptive gate when vision returns. The lower panel aggregates 500 Object-suite episodes aligned at blackout onset.

TABLE V

TRANSIENT VISUAL OOD (CAMERA BLACKOUT). DUET HOLDS THE LAST RELIABLE ABSOLUTE WAYPOINT. STEP-15 MEDIUM USES 2000 PAIRED EPISODES; STEP-23 AGGREGATES EIGHT SUITE $\times$ SEVERITY GROUPS.

Protocol	Suite	DUET-Rel	DUET	Δ	p
<i>Step 15, medium (N=500)</i>					
	Spatial	91.2	96.8	+5.6	3×10^{-5}
	Object	97.4	97.0	-0.4	0.84
	Goal	80.8	88.2	+7.4	5×10^{-4}
	LIBERO-10	64.2	86.6	+22.4	1×10^{-17}
	Overall	83.4	92.2	+8.8	—
<i>Step 23, medium+severe (N=4000)</i>					
	Overall	73.5	90.8	+17.3	—

live gate afterward. With this hold, Table V shows 92.2% versus 83.4% on the matched step-15 medium protocol (+8.8 pp). The gain is largest on LIBERO-10 (+22.4 pp), where DUET-Rel loses the long-horizon plan during dropout; Object is already at ceiling. A longer step-23 dropout widens the gap to +17.3 pp (90.8% vs. 73.5%). The remaining failure mode is Goal “open the top drawer and put the bowl inside” (−32 pp at step 15): a cached pre-dropout waypoint cannot finish a contact sequence that still requires vision after the screen returns.

D. Ablation Studies

Table VII separates inference-time and training-time ablations. Inference variants reuse the same dual-head checkpoint: DUET-Rel fixes $\alpha = 0$, absolute-only fixes $\alpha = 1$, fixed fusion sets $\alpha = 0.5$, and no adaptive CFG holds the absolute guidance scale at one. These comparisons test whether performance comes from adaptive switching rather than merely adding a second prediction.

Training ablations require independent checkpoints with the same data and optimization budget. RVQ-to-MLP tests discrete manifold anchoring; symmetric augmentation allows the relative head to train on perturbed images; and state-aware ABS gives the absolute branch direct proprioception,

testing whether state exclusion prevents a shortcut. No completed runs currently support these rows, so the table records the preregistered comparisons rather than expected outcomes. We evaluate components on clean execution and representative Robot-state and Action-execution shifts; the World-state drop is not used to establish recovery quality because the waypoint itself may be invalid.

E. Summary of Findings

Q1 remains open: the external reported rows establish the standard LIBERO range, but the matched DUET and DUET-Rel clean benchmark is still pending. **Q2 is supported across the evaluated shifts:** DUET improves robot-state teleport by 21.7 pp, vertical World-state drop by 6.0 pp, action-execution shift by 36.7 pp, cached visual dropout by 8.8 to 17.3 pp, and the separate external-force stress test by 3.73 pp (Table VI). **Q3 remains open:** Table VII defines the required causal ablations, but their results must be supplied before component-level claims are made.

V. LIMITATIONS

DUET assumes access to absolute state labels aligned with the relative action chunks. This is natural in LIBERO and many robot logs, but may require additional calibration or state estimation on real hardware. The absolute waypoint is an RVQ demonstration anchor, not an object tracker: a World-state drop can invalidate the grasp reference even when the robot remains nearby. The bound-violation term S_t is uninformative on LIBERO, so robot-state gating relies on the residual to the predicted waypoint; this still cannot distinguish “the arm left a valid target” from “the target itself moved.” Transient visual robustness requires holding the last reliable waypoint rather than running the dual heads on corrupted frames. The matched clean benchmark, the vertical-drop Object rerun, all component ablations, and real-robot evaluation are not yet complete in this draft.

VI. CONCLUSION

We presented DUET, a dual-branch VLA policy for endogenous failure detection and recovery. By aligning a local relative-action tangent field with a stateless, RVQ-anchored absolute-waypoint manifold, DUET uses cross-space disagreement to detect when execution becomes OOD and to activate a closed-loop recovery direction. This unifies failure detection and recovery without an external detector or additional recovery demonstrations.

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TABLE VI

OVERALL TASK SUCCESS UNDER THE MATCHED OOD PROTOCOLS REPORTED IN THIS SECTION. EACH ROW AGGREGATES ONLY ITS STATED SCOPE; HETEROGENEOUS SEVERITIES AND HORIZONS ARE NOT POOLED INTO A SINGLE GRAND AVERAGE.

Family	Protocol	Scope	Pairs	DUET-Rel	DUET	Δ (pp)
Robot-State	EEF teleport	4 suites, medium	2000	40.0	61.7	+21.7
World-State	Vertical grasp drop	Object, medium	500	89.2	95.2	+6.0
Action-Execution	Action perturbation	4 suites, medium+severe	4000	27.9	64.6	+36.7
Interaction stress	External force	4 suites, medium+severe	4000	92.22	95.95	+3.73
Transient Visual	Camera blackout	4 suites, step 15 medium	2000	83.4	92.2	+8.8
Transient Visual	Camera blackout	4 suites, step 23 medium+severe	4000	73.5	90.8	+17.3

TABLE VII

PLANNED COMPONENT ABLATIONS. EVERY ENTRY MARKED TODO REQUIRES A FULL MATCHED RUN; NO CONCLUSION IS INFERRED FROM AN UNTRAINED OR TEST-TIME-MISMATCHED SUBSTITUTE.

Type	Variant	Checkpoint	Clean	Robot-state	Action-execution	Isolates
Inference	Full DUET	shared	TODO	TODO	TODO	Complete controller
	DUET-Rel ($\alpha = 0$)	shared	TODO	TODO	TODO	Absolute recovery branch
	Absolute-only ($\alpha = 1$)	shared	TODO	TODO	TODO	Cost of losing local control
	Fixed fusion ($\alpha = 0.5$)	shared	TODO	TODO	TODO	Adaptive switching
	No adaptive CFG	shared	TODO	TODO	TODO	Marginal fallback
Training	RVQ $\rightarrow$ MLP	retrained	TODO	TODO	TODO	Discrete anchoring
	Symmetric augmentation	retrained	TODO	TODO	TODO	Disagreement preservation
	State-aware ABS	retrained	TODO	TODO	TODO	State-free global reference

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